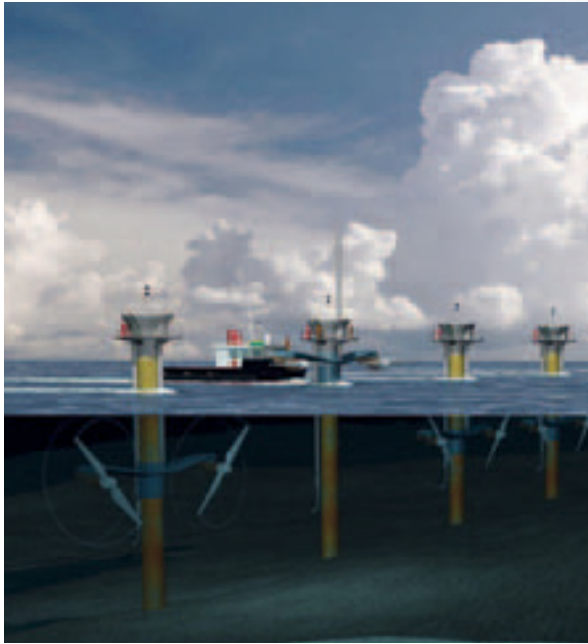


TIDAL BARRAGES ►

The tide can be harnessed by building a barrage across a river estuary, which lets the water flood in freely, then releasing it through turbines that drive electricity generators. This tidal power plant in St. Malo in France has been running since 1967. Bigger programs are planned, but they could seriously damage local marine habitats.



▲ OCEAN CURRENTS

The relentless power of ocean currents has never been used to generate electricity, but there are plans to exploit the energy of the Gulf Stream where it flows between Florida and the Bahamas. This artist's impression shows how it might be done, using a series of submerged, tethered turbines to drive the generators. This could produce as much electricity as a nuclear power plant.



▲ REINVENTING THE SAIL

Provided they are used to carry heavy cargo, ships are an extremely efficient form of transportation. Even normal cargo ships burn a lot less fuel than air freight, but their efficiency could be increased by using the wind as a power source. One design uses fiberglass aerofoils that work like sails, while another uses a giant inflatable kite.

